Shallow targets under deep logistic posteriors

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Abstract

Consider the standard logistic activation, augmented Boolean inputs, no internal biases, and a linear readout. We study the actual Gaussian posterior conditioned on absolute output margin greater than 1/2 on the entire cube and exact training labels. Every bounded-weight one-hidden-layer target has polynomial sample complexity at every finite learner width and depth in the strong-prior limit. The same holds at every fixed variance exponent $\gamma > 1$. A complementary-pair construction transmits the shallow computation at energy $16L + O(\log T + 1)$ once $L - 1 \geq \log T$, matching the leading nonconstant energy lower bound. Centering realized activations and subtracting typical Gaussian matrices then reduces the finite-scale posterior to a common class of small-product Lipschitz networks. We prove the required finite class bound by elementary layer compression. The theorem does not cover the critical exponent $\gamma = 1$ or arbitrary deep teachers, and does not settle the original tanh question.

1 Model and statement

Let $X_d = \{-1, 1\}^d$, $d \geq 1$, and $\mathsf{s}(a) = (1 + e^{-a})^{-1}$. A learner has $L \geq 2$ matrices, $L - 1$ hidden layers of common width N, input $(1, x)$, and a linear scalar output $z_W(x)$. There are no internal biases. The independent Gaussian entry variances are

$$t^2/d, \qquad t^2/N, \qquad t^2/N^2$$

for the first matrix, internal matrices, and readout. We allow a prescribed architecture-dependent scale $t = t(d, L, N)$ with $0 < t \leq 1$; no constancy across widths is assumed. For each fixed $\zeta > 0$, the finite-scale theorem covers every choice $t(d, L, N) \leq N^{-\zeta/2}$, including $t(N) = N^{(1-\gamma)/2}$ for each fixed $\gamma = 1 + \zeta > 1$. For a target $f : X_d \to \{-1, 1\}$ and sample $S = ((x_i, f(x_i)))_{i=1}^n$, write

$$E_S = \{|z_W(x)| > 1/2 \text{ for every } x \in X_d\} \cap \{\operatorname{sign} z_W(x_i) = f(x_i) \text{ for every } i\}, \qquad Q_S = \mu(\cdot \mid E_S).$$

Thus both conditioning operations are retained. For iid uniform cube examples define

$$e(S) = 2^{-d} \#\{x : Q_S(\operatorname{sign} z_W(x) = f(x)) \leq 2/3\}.$$

A shallow teacher has one hidden layer of width C, all physical weights bounded by $B \geq 1$, and $f(x) z_*(x) \geq 1$ everywhere. A fixed positive teacher margin can instead be normalized by rescaling its readout.

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Theorem 1.1. *There is a universal polynomial P_0 with the following property. Every shallow teacher as above is learned, at every $N \geq C$ and $L \geq 2$, by every limiting sign posterior as $t \downarrow 0$, with*

$$n \geq P_0(d, C, B, L, \varepsilon^{-1}, \log(2/\delta)) \quad \Longrightarrow \quad \mathbb{P}_S\{e(S) > \varepsilon\} \leq \delta.$$

Here a limiting sign posterior is any subsequential limit on the finite set of Boolean truth tables, for each fixed finite architecture. Furthermore, for every fixed $\zeta > 0$ there is a polynomial P_ζ giving the same conclusion for the actual finite-scale posterior, simultaneously for all

$$N \geq C, \quad L \geq 2, \quad 0 < t \leq N^{-\zeta/2}.$$

Equivalently, the common variance in the isotropic coordinates below is at most $N^{-(1+\zeta)}$. Polynomial degrees may depend on ζ.

The theorem includes both-label samples. Its proof does not use the prior mass of constant-sign truth tables as a denominator. Constant targets are included, but are not the reason for the upper bound.

Put

$$A_1 = \sqrt{d/N}\, W_1, \qquad A_j = W_j \ (2 \leq j < L), \qquad A_L = \sqrt{N}\, v, \qquad \kappa(A) = \sum_{j=1}^{L} \|A_j\|_F^2 .$$

All parameter coordinates now have variance $\sigma^2 = t^2/N$. The input is $u_0 = (1, x)/\sqrt{d}$, the normalized hidden states are $u_j = h_j/\sqrt{N}$, and the scalar hidden activation is $\mathsf{s}(\sqrt{N}a)/\sqrt{N}$. It is $1/4$-Lipschitz, and $\|u_j\| \leq 1$ for every hidden layer.

2 A width-independent finite class bound

We need only a bound polynomial in depth; no circuit-compression theorem is used. The activations in this section may differ between neurons.

Lemma 2.1. *Fix a finite set $X \subset \mathbb{R}^D$ of radius at most two, with $\log(2|X|) \leq d+1$, and an integer depth L. Consider networks whose hidden scalar activations fix zero and are 1-Lipschitz, whose product of matrix Frobenius norms is at most $P \geq 1$, and whose final additive scalar offset has magnitude at most $2P$. The sign functions with absolute margin at least $3/8$ on X belong to a finite class $\mathcal{F}$ such that*

$$\log |\mathcal{F}| \leq \mathrm{poly}(D, d, L, P).$$

The class is common to all original widths, activation maps, and real parameter values.

Proof. We first compress one hidden layer. Let W, V be its incoming and outgoing matrices; let w_i be row i of W, v_i column i of V, and set $a_i = \|w_i\|$, $b_i = \|v_i\|$, $A = \sum_i a_i b_i \leq \|W\|_F \, \|V\|_F$. Omit zero terms. Replacing ϕ_i by $\psi_i(s) = \phi_i(a_i s)/a_i$ and the incoming row by w_i/a_i gives

$$V\phi(Wu) = \sum_i a_i v_i \psi_i((w_i/a_i)u).$$

Sample m independent indices with probabilities $a_i b_i / A$. For each sample use incoming row w_i/a_i, activation ψ_i, and outgoing column $Av_i/(mb_i)$. Their incoming and outgoing Frobenius norms are exactly $\sqrt{m}$ and $A/\sqrt{m}$, so the product does not increase. The output is an unbiased average of vectors of norm at most $A\,\|u\|$.

If S is the product of preceding norms, the current states on X have norm at most $2S$. The expected norm error of the average is at most $2AS/\sqrt{m}$. Bounded-difference concentration, followed by a union bound over X, gives a choice of the sampled indices with preactivation error at most $CAS\sqrt{(d+1)/m}$ simultaneously on X. The suffix is Lipschitz with constant at most the product of its matrix norms. Hence the final scalar error is at most $CP\sqrt{(d+1)/m}$, irrespective of the original layer widths. This also holds when V is the final readout. Compress every hidden layer in succession, using

$$m = \left\lceil CP^2L^2(d+1) \right\rceil .$$

The total output error is at most $1/16$, after increasing C, and every hidden width is at most m. Every compression preserves zero-fixing 1-Lipschitz activations and does not increase the global product.

Now normalize each hidden matrix to Frobenius norm one. If its original norms are $a_1, \dots, a_{L-1}$, divide the hidden state at layer j by $S_j = \prod_{i \le j} a_i$, replace its scalar activation by $\phi(S_j s)/S_j$, and multiply the final readout by S_{L-1}. All activations remain zero-fixing and 1-Lipschitz; the readout norm is at most P, and all hidden states have norm at most two. If a hidden matrix has norm zero, the network is constant and can be listed separately.

Round hidden matrices in Frobenius norm to accuracy $\eta = c/(PL)$, and round the readout and final offset to accuracy c. Store every scalar activation on a uniform grid of spacing $\eta' = c/(PL\sqrt{m})$ in $[-7, 7]$, rounding its values to this accuracy and using linear interpolation. The table differs from the original activation by at most $2\eta'$ on that interval. The elementary state error recurrence, using the original activation's Lipschitz constant one, shows that all rounded states have norm below three and the final output changes by at most $1/16$, for a sufficiently small universal c. For example the accumulated hidden error is bounded by $CL(3\eta + \sqrt{m}\eta')$, since the rounded matrix norms are at most $1 + \eta$. Readout and offset errors can be made smaller fixed constants by reducing c. All preactivations stay in the stored interval.

There are polynomially many matrix entries and table entries in D, d, L, P, each with polynomial logarithmic range and precision. Thus the total description length is polynomial in these parameters. The two approximations change every output by less than $1/8$, preserving the signs of the stated margin class. Listing all descriptions proves the lemma; no precision hypothesis on the original activations is needed. □

3 A matching-energy complementary-pair wire

A nonconstant globally margined classifier has opposite labels on some cube edge. The normalized inputs at its endpoints have distance $2/\sqrt{d}$, while the outputs differ by more than one. Since logistic has derivative at most $1/4$, AM–GM gives the deterministic bound

$$\prod_{j=1}^{L} \|A_j\|_F > 4^{L-1}\sqrt{d}/2, \qquad \kappa(A) \ge 16L \exp\bigl((\log d - \log 64)/L\bigr). \tag{1}$$

In particular the leading energy coefficient is 16. This is not a finite-temperature posterior rate calculation.

Lemma 3.1. *For $D > 0$ and $m \ge 2$, put $D' = \max(D, 4)$ and $q = (D'/4)^{1/(2m)}$. There are positive weights $\alpha, b_2, \dots, b_{m-1}, c$ such that*

$$r_1 = \tanh \alpha, \quad r_s = \tanh(b_s r_{s-1}), \quad c r_{m-1} = 1/2,$$

and

$$D\alpha^2 + \sum_{s=2}^{m-1} b_s^2 + c^2 \le (m+12)q^2, \qquad c = q.$$

Moreover every consecutive product of the b_s is at most $Cm\sqrt{D'/4}$.

Proof. Set $k_s = \min(s, m-s)$ and $r_s = q^s/(\sqrt{D'}k_s)$. Then $r_s \le 1/(2qk_s) \le 1/2$ and $r_{m-1} = 1/(2q)$. Take $\alpha = \operatorname{atanh} r_1$, $b_s = \operatorname{atanh}(r_s)/r_{s-1}$, and $c = q$. For $0 < u \le 1/2$, the power series gives $(\operatorname{atanh}(u)/u)^2 \le 1 + u^2$. The reciprocal pairs in $R_s = k_{s-1}/k_s$ obey

$$((k+1)/k)^2 + (k/(k+1))^2 - 2 \le 4/k^2.$$

Consequently $\sum R_s^2 \le m - 2 + 7$, $R_s \le 2$, and $\sum k_s^{-2} < 4$. These imply

$$D\alpha^2 \le q^2 + 1/4, \qquad \sum b_s^2 \le q^2(m-2+7) + 4, \qquad c^2 = q^2,$$

whose sum is at most $(m+12)q^2$. Empty sums when $m = 2$ cause no change. Finally

$$b_s = q\frac{k_{s-1}}{k_s}\frac{\operatorname{atanh} r_s}{r_s}, \qquad \sum_s r_s^2 \le 1.$$

The k-ratios telescope, $q^m = \sqrt{D'/4}$, and the remaining factors have product at most $\exp(\frac{1}{2}\sum r_s^2)$. This proves the partial product bound. □

Let $g(x) = u \cdot h_1(x)$ be a shallow witness with signed margin one, first energy E_1, and weighted readout energy $D = N\|u\|^2$. For $N \ge 2$, use $M = 2\lfloor N/2 \rfloor$ active neurons in each added hidden layer, split into equal groups. Let a have entries $2/M$ and $-2/M$ on the groups and zero on any surplus coordinate. Then $\|a\|^2 = 4/M$, and

$$\mathsf{s}(2z) - \mathsf{s}(-2z) = \tanh z.$$

Apply Lemma 3.1 with first coefficient $D/4$ and $m = L - 1$. The first added layer has rows $\pm 2\alpha u$ and energy at most $4D\alpha^2$. Each later added layer has rows $\pm 2b_s a^T$ and energy $16b_s^2$. The final readout $2ca$ has weighted energy $16(N/M)c^2 \le 32c^2$. Monotonicity of the scalar chain gives global signed output at least one. Thus

$$\kappa(A_*) \le E_1 + 16(L+12)\max(1, D/16)^{1/(L-1)}. \tag{2}$$

Odd widths have only an endpoint cost; they do not increase the per-layer coefficient 16.

Choose an integer

$$M_0 \ge \max\{C, 2, d+1, dC(d+1)B^2\}$$

of polynomial size. Zero-padding the teacher to width M_0 makes its first energy at most one. Balanced cloning to any $N \ge M_0$ gives $E_1 \le 2$ and $D \le D_0 := 2M_0CB^2$: each replication count lies between $N/(2M_0)$ and $2N/M_0$. Put $T = \max(1, D_0/16)$. The construction gives

$$K_L = 2 + 16(L+12)T^{1/(L-1)}, \qquad \kappa(A_*) \le K_L. \tag{3}$$

In particular, for every $L \ge 3$,

$$4\left(\frac{K_L+1}{16L}\right)^{L/2} \le 4e^{13/2}T^{3/4}. \tag{4}$$

Indeed $(K_L+1)/16 \le (L+13)T^{1/(L-1)}$ and $L/(2(L-1)) \le 3/4$. For $L = 2$ the cloned shallow witness directly has energy at most $2 + D_0$, which gives a polynomial bound as well.

Lemma 3.2. *The above witness can be chosen so that perturbing every parameter block in operator norm by at most η changes its output on the entire cube by at most $CL^2T\eta$. The same bound, with an enlarged constant, holds for the direct $L = 2$ witness.*

Proof. Use the upper bound D_0 in the scalar lemma. The first added layer's normalized Lipschitz gain $\|A_2^*\|_{\mathrm{op}}/4$ is at most Cq; the last readout norm is at most Cq. The first teacher matrix has norm at most $\sqrt{2}$. The intervening gains are exactly b_s in operator norm, and Lemma 3.1 bounds every partial product by $CL\sqrt{T}$. Including the endpoint factors gives a bound CLT for every suffix affecting an injected error.

Normalized logistic hidden states of both the witness and the perturbed network have norm at most one. Therefore their state differences satisfy

$$e_j \le (\left\|A_j^*\right\|_{\mathrm{op}}/4)e_{j-1} + \eta/4,$$

with first forcing at most $\sqrt{2}\eta/4$. Summing the L forced errors, and treating the final linear readout similarly, proves the result. The proof uses the explicit partial products; total energy alone would not supply this polynomial stability bound. □

4 Centering and finite-scale Gaussian subtraction

Fix an anchor $x_0 \in X_d$. Center every normalized hidden state at that anchor. Writing $b_{j,i} = (A_j u_{j-1}(x_0))_i$, the centered scalar maps are

$$a_{b,N}(r) = \frac{\mathsf{s}(\sqrt{N}(b+4r)) - \mathsf{s}(\sqrt{N}b)}{\sqrt{N}}.$$

They fix zero and are 1-Lipschitz. The centered linear blocks are $A_j/4$ for hidden layers and A_L for the readout. Thus the centered product is at most

$$P(A) = 4^{-(L-1)} \prod_j \|A_j\|_F \le 4\left(\frac{\kappa(A)}{16L}\right)^{L/2}. \tag{5}$$

The centered input $(0, x - x_0)/\sqrt{d}$ has norm at most two. The output is $z_A(x_0)$ plus this centered computation. If the truth table is nonconstant, an opposite-sign point implies

$$|z_A(x_0)| \le 2P(A). \tag{6}$$

Constants can be listed separately.

For $N \ge M_0$, let $\mathcal{T}$ be the intersection of the operator-norm balls of radius tb in all parameter blocks, where

$$b^2 = 16\log 9 + 8\log(8L).$$

This is a closed convex centrally symmetric set with $\mu(\mathcal{T}) \ge 1/2$. For completeness, a $1/4$-net in each unit sphere has cardinality at most 9^r in dimension r. The bilinear Gaussian tail and the net bound give $\mathbb{P}(\|G\|_{\mathrm{op}} > tb) \le 2\,9^{r+c}\exp(-Nb^2/8)$ for an $r \times c$ block of variance t^2/N. Here $r + c \le 2N$, since $N \ge d + 1$, so a union bound over the L blocks proves the claim.

Assume

$$tb \le \xi := c/(L^2T) \tag{7}$$

for a sufficiently small universal c. Lemma 3.2 puts $A_* + \mathcal{T}$ inside global signed margin $3/4$, hence inside every E_S labeled by the target. Gaussian translation and symmetry give

$$\mu(E_S) \geq \mu(A_* + \mathcal{T}) \geq \tfrac{1}{2} \exp[-K_L/(2\sigma^2)]. \tag{8}$$

Indeed the translated density is the original density multiplied by $\exp[-\|A_*\|^2/(2\sigma^2) - \langle A_*, A\rangle/\sigma^2]$; symmetry replaces the last factor by a hyperbolic cosine of mean at least one. Gaussian isoperimetry for a set of mass at least $1/2$ gives $\mu\{\mathrm{dist}(A, \mathcal{T}) > r\} \leq \exp[-r^2/(2\sigma^2)]$. Dividing by (8) proves the actual posterior estimate

$$Q_S\{\mathrm{dist}(A, \mathcal{T})^2 > K_L + 1\} \leq 2\exp[-1/(2\sigma^2)]. \tag{9}$$

Only this standard Gaussian isoperimetric inequality is used as a probabilistic input beyond elementary concentration.

On the complementary event, project A onto $\mathcal{T}$ and write $A = U + B$, where $\kappa(U) \leq K_L + 1$ and every block of B has operator norm at most tb. Center the *realized* network A at x_0 and freeze its exact maps $a_{b,N}$. Replace its centered linear matrices by those of U, keeping both the frozen maps and the actual anchor output. This surrogate need not be a native logistic network. It belongs to the larger Lipschitz class of Lemma 2.1, which is all that counting requires. The anchor maps are not recomputed after subtraction.

To bound the change, divide all U blocks, including the readout, by four and place a factor four outside the final output. Set $K' = (K_L + 1)/16$ and $\eta = tb/4$. A telescoping replacement of linear blocks, using zero-preserving 1-Lipschitz maps and input radius two, gives error at most

$$8L\eta \left(\frac{(\sqrt{K'} + \sqrt{L}\eta)^2}{L - 1} \right)^{(L-1)/2}. \tag{10}$$

To verify it, each of the L terms is bounded by 8η times the product of the other $L - 1$ block norms, each enlarged by at most η. Their squared sum is at most $(\sqrt{K'} + \sqrt{L}\eta)^2$; apply AM–GM. For $L \geq 3$, (7) and $K' \leq (L + 13)T^{1/(L-1)}$ bound the displayed power by $C\sqrt{T}$. Indeed the added energy is bounded by a universal constant, after which the ratio is at most $(1 + C/(L - 1))T^{1/(L-1)}$. Thus (10) is at most $1/8$, after decreasing c. For $L = 2$, use its direct bound $K' \leq C(1 + T)$ and the same conservative condition (7).

The surrogate preserves every sign of the globally margined posterior network and has margin greater than $3/8$. Its product is polynomial in T by (4). Since every frozen map fixes zero, its centered computation vanishes at the anchor. If its truth table is nonconstant, the retained offset is automatically at most twice its own product, by the same opposite-sign argument as (6). Lemma 2.1 therefore supplies one class $\mathcal{F}$ with

$$\log|\mathcal{F}| \leq \mathrm{poly}(d, C, B, L), \qquad Q_S\{\mathrm{sign}\, z_A \notin \mathcal{F}\} \leq 2e^{-1/(2\sigma^2)}. \tag{11}$$

The class is independent of the sample, width, and realized anchor values.

5 Small widths, cold limits, and statistical conversion

We record the elementary finite-width estimate needed below the large-width threshold. It also avoids any appeal to sigmoid VC theory.

Lemma 5.1. *Suppose a width-N, depth-L logistic target witness has signed margin one and isotropic energy at most $K_0 \geq 1$. For $0 < t \leq 1$, its posterior has sufficient sample size polynomial in*

$$d, N, L, K_0, \varepsilon^{-1}, \log(2/\delta),$$

uniformly in t.

Proof. The parameter dimension is $p = N(d+2) + (L-2)N^2$. On an energy ball $\kappa \leq R$, the map from parameters to any cube output has a Lipschitz constant Λ_R with

$$\log(2 + \Lambda_R) \leq CL \log(2 + R + L).$$

For example differentiate layer by layer: all hidden states have norm at most one, the input norm is at most $\sqrt{2}$, each derivative is bounded by $1/4$, and every downstream block norm is at most $\sqrt{R}$. This proves a bound $CL \max(1, \sqrt{R})^L$, which suffices.

Let $K = 2K_0 + 2$ and $r = \min(1, 1/(4\Lambda_K))$. The ball of radius r around the witness has signed margin at least $3/4$ and energy at most K. Its inscribed cube with half-side $\min(r/\sqrt{p}, \sigma)$ gives

$$\mu(E_S) \geq \exp[-K/(2\sigma^2) - pT_0], \qquad T_0 = \log\left(C \max\{1, \sqrt{p/N}/r\}\right).$$

Here the clipping at σ is essential: since $\sigma \leq 1/\sqrt{N}$, the volume-to-Gaussian-scale ratio introduces no $\log(1/t)$ term. The isotropic Gaussian tail $\mu\{\kappa > R\} \leq 2^{p/2} e^{-R/(4\sigma^2)}$ shows that the posterior mass outside

$$R_\beta = 2K + \frac{4}{N}\{p(T_0 + \tfrac{1}{2}\log 2) + \log(1/\beta)\}$$

is at most β. This radius is polynomial in the displayed resources and $\log(1/\beta)$.

A parameter net of mesh $1/(8\Lambda_{R_\beta+1})$ in that ball has log-cardinality polynomial in the same resources. Every globally margined network in the ball has exactly the signs of a net representative, since the output approximation error is below $1/4$. Thus a common finite class contains posterior mass at least $1 - \beta$. The finite-class argument given below completes the proof. □

For $C \leq N < M_0$ and $N \geq 2$, zero-padding the shallow teacher to width N gives first energy at most $dC(d+1)B^2/N$ and readout energy at most NCB^2. For $L \geq 3$, applying (2) gives a witness energy polynomial in d, C, B, L, N; for $L = 2$, use the direct shallow witness. Hence Lemma 5.1 handles these widths. For $N = 1$ every logistic truth table is constant, since the last hidden value is positive and the readout is scalar; one example fixes its sign everywhere.

For the finite-scale part of Theorem 1.1, fix $\zeta > 0$. There is a threshold $N_* \geq M_0$, polynomial in $d, C, B, L, \log(2/\beta)$ with degree depending only on ζ, such that every $N \geq N_*$ satisfies both (7) and $2e^{-1/(2\sigma^2)} \leq \beta$ whenever $t \leq N^{-\zeta/2}$. This follows from $b = O(\sqrt{\log(2L)})$, $T = \mathrm{poly}(d, C, B)$, and $\sigma^2 \leq N^{-(1+\zeta)}$. Equation (11) handles all widths above N_*. For widths below it, the just-described witness and Lemma 5.1 give a polynomial bound because N_* itself is polynomial. Take the union of the large-width class with the finite parameter-net classes for the integers $C \leq N < N_*$. Its log-cardinality increases by at most $\log(N_* + 1)$ beyond the largest individual bound, and remains polynomial. This gives one sample event for all widths and eligible scales at the fixed depth.

For the cold statement, at every fixed finite architecture a strict-margin witness of energy at most K implies

$$Q_S\{\kappa > K + \rho\} \longrightarrow 0 \qquad (\sigma \downarrow 0) \tag{12}$$

for each $\rho > 0$, uniformly over target-labeled samples. A sufficiently small fixed neighborhood of the witness retains its signs and margin and has energy at most $K + \rho/2$. Its Gaussian mass is

bounded below by a constant times $\sigma^{-p}e^{-(K+\rho/2)/(2\sigma^2)}$. The Gaussian energy tail above $K+\rho$ is at most a polynomial in σ^{-1} times $e^{-(K+\rho)/(2\sigma^2)}$, so their ratio tends to zero. With (3), (5), and Lemma 2.1, every limiting sign posterior at $N \geq M_0$ is supported on a common class of polynomial log-cardinality. The preceding finite-width argument covers the remaining widths uniformly in the cold limit. Passing Gibbs error through a sign-posterior limit is valid on the finite truth-table simplex; no continuity of the inclusive confidence threshold is assumed.

Finally, let a sample-independent class of size M contain posterior mass at least $1-\beta$. With probability at least $1-\delta$, every sample-consistent member of this class has true error at most α, provided

$$n \geq \frac{\log M + \log(1/\delta)}{\alpha}.$$

Every draw from Q_S is sample consistent. Its expected true error is therefore at most $\alpha+\beta$. If correct-label confidence is at most $2/3$, its wrong-label probability is at least $1/3$, so $e(S) \leq 3(\alpha+\beta)$. Taking $\alpha = \beta = \varepsilon/12$ leaves strict slack at the stated threshold. For a limiting posterior supported on the class take $\beta = 0$.

To obtain the simultaneous assertion across depths, fix the other target and statistical resources and put $\delta_L = \delta/[L(L-1)]$ for $L \geq 2$. At each depth the argument above already gives one event for all widths and eligible scales. For a fixed sample size n, union bound over the depths whose sufficient sample condition is met, using δ_L in place of δ. Their failure probabilities sum to at most δ. Since $\log(2/\delta_L) = \log(2/\delta) + \log(L(L-1))$, enlarging the existential polynomial sample budget absorbs this additional term. The same reasoning applies to the cold classes. This proves the stated simultaneity without asserting a depth-uniform positive cold threshold, and completes Theorem 1.1.

6 Scope and a concrete shallow example

Parity is included in the shallow-target hypothesis. To see this directly, put $k(x) = (d + \sum_i x_i)/2 \in \{0, \ldots, d\}$ and $y(k) = (-1)^k$. This is full parity up to the fixed sign $(-1)^d$, which can be absorbed into the readout. The step expansion

$$y(k) = y(0) + \sum_{j=1}^{d} [y(j) - y(j-1)]\, \mathbf{1}_{\{k > j-1/2\}}$$

is approximated on these integer values by replacing each step with $\mathsf{s}(A(k-j+1/2))$, where $A = 2\log(16d)$. The total approximation error is at most $2de^{-A/2} \leq 1/8$. The constant term is realized by a zero-weight hidden unit with output $1/2$. Multiplying the final output by two gives signed margin at least one. There are $d+1$ hidden units and weights bounded by $O(d\log(2d))$.

Thus neither parity nor another bounded-weight shallow logistic target can yield an exponential-in-depth sample lower bound in the regimes of Theorem 1.1. The proof leaves the exact critical exponent $\gamma = 1$, and a fixed positive canonical prefactor t, open: then the typical Gaussian block size tb need not satisfy (7). It also leaves arbitrary deep teachers open. The offset of standard logistic is handled here by centering and complementary transmission; this result does not assert the analogous theorem for other activations or resolve the original tanh problem.

References

[1] C. Borell, *The Brunn–Minkowski inequality in Gauss space*, Inventiones Mathematicae **30** (1975), 207–216.